Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases
to guide strategic business decisions

3,900	$59.76	3.75 / 5	10
Customers	Avg Purchase	Avg Rating	SQL Queries

Python · Pandas | **MySQL · SQL** | **Power BI**

Data Analytics Project Report

EDA · SQL Analysis · Business Intelligence

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1. Business Problem Statement

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. Management has noticed changes in purchasing patterns across demographics, product categories, and sales channels. They are particularly interested in uncovering which factors — discounts, reviews, seasons, or payment preferences — drive consumer decisions and repeat purchases.

Project Deliverables

01 Data Preparation & Modeling (Python)

Clean and transform the raw dataset — handle missing values, engineer features (age groups, purchase frequency), standardize columns, and load into the database.

02 Data Analysis (SQL)

Organize data into a structured format, simulate business transactions, and run 10 queries to extract insights on customer segments, loyalty, and purchase drivers.

03 Visualization & Insights (Power BI)

Build an interactive dashboard highlighting key patterns — revenue by category, age group, subscription status, and shipping type with dynamic slicers.

04 Report & Presentation

Write a clear project report summarizing key findings and business recommendations. Prepare a presentation that communicates insights to stakeholders.

05 GitHub Repository

Include all Python scripts, SQL queries, Power BI files, and documentation in a well-structured repository for reproducibility.

2. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The analysis leverages **Python (pandas)** for data cleaning and feature engineering, **MySQL** for structured querying, and **Power BI** for interactive dashboards.

3,900	$59.76	3.75 / 5	18	4
Total Customers	Avg Purchase	Avg Rating	Dataset Columns	Product Categories

3. Dataset Overview

The dataset contains 3,900 rows and 18 columns covering customer demographics (age, gender, location), purchase details (item, category, amount), and behavioral attributes (subscription status, frequency, discount usage, review ratings). Only the **Review Rating** column contained 37 missing values.

Column	Description	Type
customer_id	Unique customer identifier	Integer
age	Customer age (18–70)	Integer
gender	Male / Female	Categorical
item_purchased	Product name purchased	String
category	Clothing / Accessories / Footwear / Outerwear	Categorical
purchase_amount	Transaction value in USD ($20–$100)	Float
review_rating	Rating 1–5 (37 missing — imputed)	Float
subscription_status	Yes / No	Categorical
shipping_type	Express / Standard / Free / 2-Day / etc.	Categorical
location	US State of purchase (e.g. Montana, California)	String
size	Clothing/Footwear size (S / M / L / XL)	Categorical
color	Product color (e.g. Black, Blue, Red)	String
season	Purchase season (Spring / Summer / Fall / Winter)	Categorical
payment_method	Credit Card / PayPal / Cash / etc.	Categorical
discount_applied	Discount applied? Yes/No (promo_code_used was identical — dropped)	Categorical
previous_purchases	Count of past purchases (1–50)	Integer
frequency_of_purchases	Weekly / Monthly / Quarterly / etc.	Categorical
age_group	Engineered: Young Adult / Adult / Middle-aged / Senior	Engineered
purchase_frequency_days	Engineered: numeric frequency in days	Engineered

4. Python — Data Preparation

The Jupyter Notebook handles all data preparation using **Python (pandas)** — loading the raw CSV, cleaning missing values, engineering new features, and exporting to MySQL. Each of the 8 steps below shows the exact code and actual output.

Step 01 Data Loading & Initial Exploration

Imported the raw CSV using pandas. Ran df.head() to preview rows, df.info() to check column types and null counts, and df.describe() for summary statistics.

```
import pandas as pd df = pd.read_csv('customer_shopping_behavior.csv') df.head()
```

```
Customer ID Age Gender Item Purchased Category Purchase Amount (USD) ...
0 1 55 Male Blouse Clothing 53 ...
1 2 19 Male Sweater Clothing 64 ...
2 3 50 Male Jeans Clothing 73 ...
3 4 21 Male Sandals Footwear 90 ...
4 5 45 Female Blouse Clothing 49 ...
```

Step 02 Structure Check — df.info()

Verified 3,900 rows and 18 columns. Identified that **Review Rating** was the only column with missing values — 3,863 non-null out of 3,900 (37 missing).

```
df.info()
```

```
RangeIndex: 3900 entries, 0 to 3899 | Total columns: 18

# Column Non-Null Count Dtype
0 Customer ID 3900 non-null int64
1 Age 3900 non-null int64
2 Gender 3900 non-null object
3 Item Purchased 3900 non-null object
4 Category 3900 non-null object
5 Purchase Amount (USD) 3900 non-null int64
6 Location 3900 non-null object
7 Size / Color / Season 3900 non-null object
10 Review Rating 3863 non-null float64 <-- 37 missing values
11 Subscription Status 3900 non-null object
12 Shipping Type 3900 non-null object
13 Discount Applied 3900 non-null object
14 Promo Code Used 3900 non-null object
15 Previous Purchases 3900 non-null int64
16 Payment Method / Frequency of Purchases 3900 non-null object
```

Step 03 Missing Value Treatment

Checked null values with df.isnull().sum(). Imputed the 37 missing Review Rating values using **group-wise median per Category** — this preserves category-level rating distributions better than a global median fill.

```
df.isnull().sum() # Impute missing Review Rating with category-wise median df['Review Rating'] =
df.groupby('Category')['Review Rating'] \ .transform(lambda x: x.fillna(x.median()))
df.isnull().sum() # Confirm all zeros
```

```
Before imputation:
Review Rating 37 (all other 17 columns: 0 missing)

After imputation:
Review Rating 0 (confirmed — no missing values remain)
```

Step 04 Column Standardization — snake_case

Renamed all columns to lowercase snake_case for SQL compatibility. Also renamed purchase_amount_(usd) to purchase_amount for cleaner querying.

```
df.columns = df.columns.str.lower() df.columns = df.columns.str.replace(' ', '_') df =
df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'}) df.columns
```

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
'purchase_amount', 'location', 'size', 'color', 'season',
'review_rating', 'subscription_status', 'shipping_type',
'discount_applied', 'promo_code_used', 'previous_purchases',
'payment_method', 'frequency_of_purchases'],
dtype='object')
```

Step 05 Feature Engineering — age_group

Created a new age_group column using pd.qcut() with 4 quantile bins. Quantile-based binning ensures **balanced group sizes** — each segment contains roughly the same number of customers.

```
labels = ['Young Adult', 'Adult', 'Middle-aged', 'Senior'] df['age_group'] = pd.qcut(df['age'],
q=4, labels=labels) df[['age', 'age_group']].head(10)
```

```
age age_group
0 55 Middle-aged
1 19 Young Adult
2 50 Middle-aged
3 21 Young Adult
4 45 Middle-aged
5 32 Adult
6 62 Senior
7 28 Young Adult
8 57 Senior
9 38 Adult
```

Step 06 Feature Engineering — purchase_frequency_days

Mapped text-based purchase frequency labels to numeric day values, enabling quantitative analysis of how often each customer shops.

```
frequency_mapping = { 'Fortnightly' : 14, 'Weekly' : 7, 'Monthly' : 30, 'Quarterly' : 90,
'Bi-Weekly' : 14, 'Annually' : 365, 'Every 3 Months': 90 } df['purchase_frequency_days'] =
df['frequency_of_purchases'].map(frequency_mapping) df[['purchase_frequency_days',
'frequency_of_purchases']].head(6)
```

```
purchase_frequency_days frequency_of_purchases
0 14 Fortnightly
1 14 Fortnightly
2 7 Weekly
3 7 Weekly
4 30 Monthly
5 90 Quarterly
```

Step 07 Data Consistency — Remove Duplicate Column

Checked whether discount_applied and promo_code_used carried identical information. The .all() check returned True — confirming 100% duplication. Dropped promo_code_used, reducing columns from 18 to 17.

```
# Verify columns are identical df[['discount_applied', 'promo_code_used']].head(5)
(df['discount_applied'] == df['promo_code_used']).all() # Drop redundant column df =
df.drop('promo_code_used', axis=1) print(f'Remaining columns: {len(df.columns)}')
```

```
discount_applied promo_code_used
0 Yes Yes
1 Yes Yes
2 Yes Yes
3 No No

True <-- columns are 100% identical

Remaining columns: 17
```

Step 08 MySQL Connection & Data Load

Connected to MySQL using SQLAlchemy with the pymysql driver. Loaded the fully cleaned and engineered DataFrame into the **customer** table inside the **customer_behavior** database using df.to_sql().

```
import mysql.connector from sqlalchemy import create_engine from sqlalchemy.engine import URL url =
URL.create( drivername = 'mysql+pymysql', username = 'root', host = 'localhost', database =
'customer_behavior' ) engine = create_engine(url) print('Connected successfully!') # Load cleaned
DataFrame into MySQL df.to_sql('customer', engine, if_exists='replace', index=False) print("Data
loaded into table 'customer' in database 'customer_behavior'")
```

```
Connected successfully!

Data successfully loaded into table 'customer' in database 'customer_behavior'
```

■ Key Preparation Decisions

- Group-wise median imputation (by Category) preserves rating distributions better than a global fill.
- Quantile binning (qcut) for age_group ensures all 4 segments have equal customer counts.
- Dropping the duplicate promo_code_used column avoids skewed analysis — both columns were 100% identical.
- Mapping frequency labels to numeric days enables quantitative purchase-frequency analysis in SQL.

5. SQL Analysis & Results

Ten SQL queries were executed against the MySQL **customer_behavior** database. Each query is shown with the SQL code and actual result table.

5.1 Revenue by Gender

Business Question: What is the total revenue generated by male vs. female customers?

```
SELECT gender,
SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;
```

gender	revenue
Female	75191
Male	157890

Table: Revenue by Gender

5.2 High-Spending Discount Users

Business Question: Which customers used a discount but spent above the average purchase amount?

```
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
AND purchase_amount > (SELECT AVG(purchase_amount) FROM customer);
```

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
...	... (839 total rows)

Table: High-Spending Discount Users (839 total rows, sample shown)

5.3 Top 5 Products by Review Rating

Business Question: Which 5 products have the highest average review rating?

```
SELECT item_purchased,
AVG(review_rating) AS avg_rating
FROM customer
GROUP BY item_purchased
ORDER BY avg_rating DESC
LIMIT 5;
```

item_purchased	avg_rating
Gloves	3.8614
Sandals	3.8444
Boots	3.8188
Hat	3.8013
Skirt	3.7848

Table: Top 5 Products by Avg Review Rating

5.4 Shipping Type Comparison

Business Question: Compare average purchase amounts between Standard and Express shipping.

```
SELECT shipping_type,
ROUND(AVG(purchase_amount), 2) AS avg_amount
FROM customer
WHERE shipping_type IN ('Express', 'Standard')
GROUP BY shipping_type;
```

shipping_type	avg_amount
Standard	58.46
Express	60.48

Table: Avg Purchase by Shipping Type

5.5 Subscribers vs. Non-Subscribers

Business Question: Do subscribed customers spend more? Compare avg spend and total revenue.

```
SELECT subscription_status,
COUNT(customer_id) AS total_customers,
ROUND(AVG(purchase_amount), 2) AS avg_spend,
SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY subscription_status;
```

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

Table: Subscriber vs. Non-Subscriber Comparison

■ Insight

- Subscribers (27% of customers = 1,053) generated $62,645 total revenue; Non-subscribers (73% = 2,847) generated $170,436.
- Avg spend is nearly identical ($59.49 vs $59.87) — non-subscribers slightly outspend per transaction.
- The subscription program needs repositioning: focus on volume retention and frequency, not just average spend incentives.

5.6 Discount-Dependent Products

Business Question: Which 5 products have the highest percentage of discounted purchases?

```
SELECT item_purchased,
ROUND(SUM(discount_applied = 'Yes') / COUNT(*) * 100, 2) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Table: Top 5 Discount-Dependent Products

5.7 Customer Segmentation

Business Question: Classify customers into New, Returning, and Loyal segments.

```
SELECT
CASE
WHEN previous_purchases = 1 THEN 'New'
WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
ELSE 'Loyal'
END AS customer_segment,
COUNT(*) AS number_of_customers
FROM customer
GROUP BY customer_segment;
```

customer_segment	number_of_customers
Loyal	3116
Returning	701
New	83

Table: Customer Segmentation Result

■ Insight

- 79.9% of customers are Loyal (>10 purchases) — an exceptionally strong base for retention-focused strategies.
- Only 2.1% (83 customers) are New — acquisition efforts should be scaled up to broaden the funnel.
- The 701 Returning customers represent the best conversion opportunity: personalized nudges can push them to Loyal status.

5.8 Top 3 Products per Category

Business Question: What are the top 3 most purchased products within each category?

```
SELECT item_purchased, category, purchase_count
FROM (
SELECT item_purchased, category,
COUNT(*) AS purchase_count,
ROW_NUMBER() OVER (
PARTITION BY category ORDER BY COUNT(*) DESC
) AS rnk
FROM customer
GROUP BY item_purchased, category
) ranked
WHERE rnk <= 3
ORDER BY category, purchase_count DESC;
```

item_purchased	category	purchase_count
Jewelry	Accessories	171
Sunglasses	Accessories	161
Belt	Accessories	161
Blouse	Clothing	171
Pants	Clothing	171
Shirt	Clothing	169
Sandals	Footwear	160
Shoes	Footwear	150
Sneakers	Footwear	145
Jacket	Outerwear	163
Coat	Outerwear	161

Table: Top 3 Products per Category

■ Insight

• Jewelry and Blouse lead their categories with 171 purchases each — prime candidates for hero banner placement.

• Footwear shows the widest spread (Sandals 160 vs Sneakers 145) — Sneakers may need a promotional push.

• Jacket dominates Outerwear (163) suggesting strong seasonal demand worth capitalizing on with early stock.

5.9 Repeat Buyers & Subscriptions

Business Question: Are customers with >5 purchases more likely to subscribe?

```
SELECT
CASE WHEN previous_purchases > 5 THEN 'Repeat Buyer'
ELSE 'Non-Repeat Buyer' END AS buyer_type,
subscription_status,
COUNT(*) AS customer_count,
ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER
(PARTITION BY
CASE WHEN previous_purchases > 5
THEN 'Repeat Buyer'
ELSE 'Non-Repeat Buyer' END), 2) AS percentage
FROM customer
GROUP BY buyer_type, subscription_status
ORDER BY buyer_type, subscription_status;
```

buyer_type	subscription_status	customer_count	percentage
Non-Repeat Buyer	No	329	77.59%
Non-Repeat Buyer	Yes	95	22.41%
Repeat Buyer	No	2,518	72.44%
Repeat Buyer	Yes	958	27.56%

Table: Repeat Buyers & Subscription Status

5.10 Revenue by Age Group

Business Question: What is the total revenue contribution of each age group?

```
SELECT age_group,
SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

Table: Revenue by Age Group

6. Power BI Dashboard

An interactive Power BI dashboard was built to present insights visually. The dashboard features slicers for Subscription Status, Gender, Category, and Shipping Type, enabling dynamic cross-filtering across all charts.

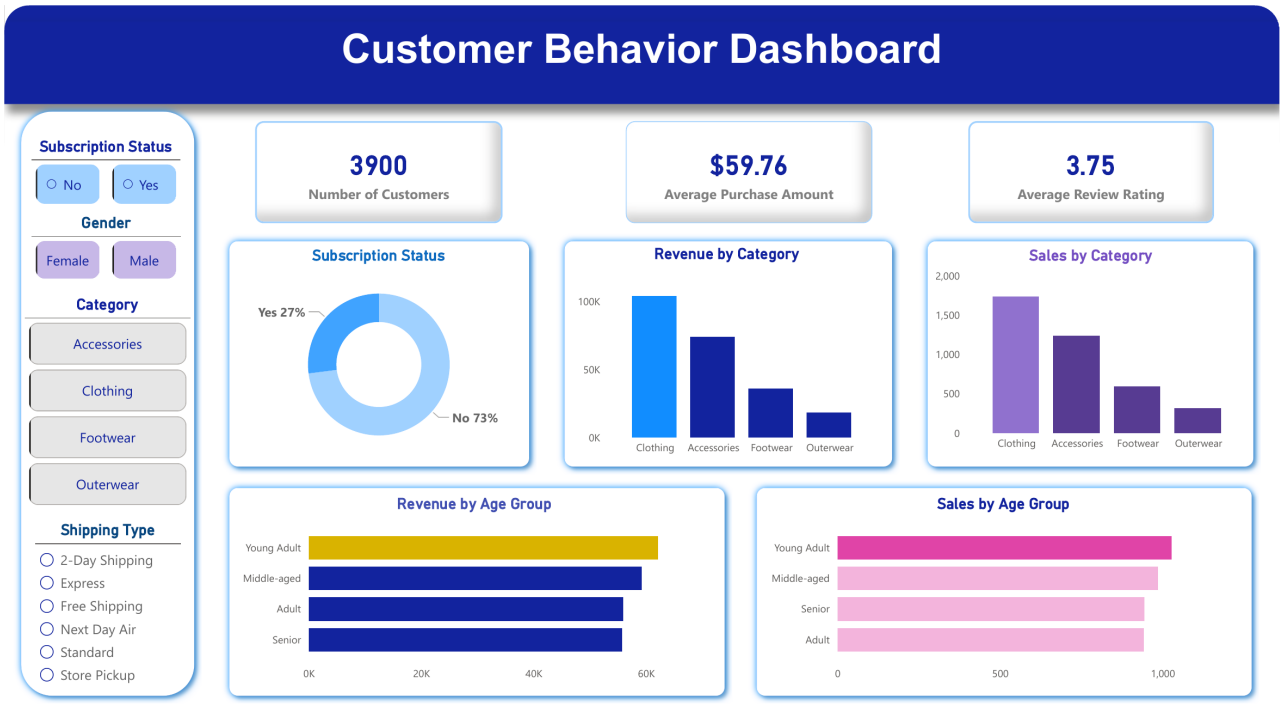

Figure 1: Customer Behavior Interactive Dashboard — Power BI Desktop

Dashboard Component	Description & Key Numbers
KPI Cards (3)	3,900 Customers · $59.76 Avg Purchase · 3.75 Avg Rating
Subscription Donut	73% No vs 27% Yes — with interactive subscription slicer
Revenue by Category	Clothing leads (~$100K+), followed by Accessories, Footwear, Outerwear
Sales by Category	Volume bars confirming Clothing (1,737 units) dominates all categories
Revenue by Age Group	Young Adult tops ($62K), all groups closely clustered between $55K–$62K
Sales by Age Group	Young Adult leads volume; Senior and Adult nearly tied for lowest
Slicers / Filters	Subscription Status · Gender · Category (4) · Shipping Type (6)

7. Key Insights & Business Recommendations

Boost Subscription Program

■ Finding:
Only 27% of customers subscribe, yet 72% of repeat buyers (3,476 customers with >5 purchases) have not subscribed. The program is severely under-leveraged given the loyal customer base.

■ Recommendation:
Target repeat buyers with a free 1-month trial. Promote exclusive benefits — early access, free shipping, member-only discounts — to convert the 2,518 non-subscribed repeat buyers.

Review Discount Policy

■ Finding:
Hat (50%), Sneakers (49.66%), and Coat (49.07%) have nearly half of all purchases discounted. Persistent always-on discounting erodes margins and trains customers to wait for deals.

■ Recommendation:
Switch to time-limited flash discounts. Test removing discounts on top-rated products (Gloves, Sandals) to protect perceived value and improve gross margins.

Activate Returning Customers

■ Finding:
701 customers are in the Returning segment — one step away from the highest Loyal tier. These are high-intent buyers who simply need a stronger nudge to return more frequently.

■ Recommendation:
Deploy a targeted email sequence after every 3rd purchase with personalized product recommendations and a "Complete your loyalty journey" incentive message.

Spotlight Top-Rated Products

■ Finding:
Gloves (3.86), Sandals (3.84), and Boots (3.82) are the highest-rated products yet are likely under-featured in current marketing given no direct correlation to top revenue.

■ Recommendation:
Feature these products in homepage hero banners, email campaigns, and social ads. Use verified star ratings as social proof on product listing and checkout pages.

Expand Customer Acquisition

■ Finding:
Only 83 customers (2.1%) are classified as New. The business has an exceptionally strong Loyal base (79.9%) but the acquisition funnel is critically narrow.

■ Recommendation:
Invest in paid acquisition channels (social ads, influencer partnerships, referral programs) to widen the top of funnel and ensure sustainable long-term revenue growth.